Introduction to AI-Assisted Research Methods and Data Analysis

Curriculum Outline

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A four-day, 24-hour introductory curriculum for clinicians and biomedical researchers. Python and R in VS Code with Jupyter notebooks; GitHub Copilot as the AI coding assistant.

Ground rule: use synthetic or fully de-identified teaching data. Never upload identifiable patient data or confidential manuscripts to public AI services.

Course at a glance

This course takes clinicians and biomedical researchers from a clinical observation to an auditable mini research report. Participants work through one continuous synthetic primary-care example, then apply the same sequence to a question of their own. The emphasis is on research alignment, transparent decisions and verified AI assistance rather than programming speed or statistical breadth.

Feature	Specification
Audience	Family medicine doctors and residents; other medical specialty residents; interested biomedical researchers
Prerequisites	No coding experience. Basic clinical or biomedical knowledge. Minimal research-methods or biostatistics knowledge is enough.
Duration	Four days, 08:30 to 16:00. Four 90-minute blocks a day; 24 instructional hours in total.
Breaks	10:00 to 10:15, lunch 11:45 to 12:45, and 14:15 to 14:30
Computing	VS Code with Jupyter notebooks. Every notebook exists in Python and in R; participants choose one. Google Colab is a no-install fallback.
AI tools	GitHub Copilot in VS Code (free plan is enough). Other assistants are welcome if participants already have them. NotebookLM for source-grounded literature work, where institutional policy allows.
Core materials	Four slide decks; five notebooks in each language with worked solutions; synthetic data and format examples; Word templates; participant workbook; facilitator guide; Quarto report template
Capstone	A two-page-equivalent mini research report, a runnable notebook, literature evidence, analysis decisions and an AI-use and verification record

Continuous teaching case. Among adults attending primary care, is medication adherence associated with systolic blood pressure, and with uncontrolled blood-pressure outcome? The outcome is defined as SBP ≥ 140 mmHg or DBP ≥ 90 mmHg.

Learning outcomes

By the end of the course, participants should be able to:

1. Frame an answerable clinical or biomedical question using PICO or PECO, and review its feasibility with FINER.
2. Write null and alternative hypotheses that match the population, exposure or intervention, outcome and intended estimand.
3. Distinguish common observational and experimental designs and state the inference each can and cannot support.
4. Draw a conceptual variable map, distinguish confounders from mediators and colliders, and use a theoretical framework without claiming untested mechanisms.
5. Plan and document a literature search, and organise references, PDF annotations, literature matrices and annotated bibliographies in Zotero.
6. Move verified sources from Zotero into NotebookLM for bounded, source-grounded synthesis, then verify every important claim against the cited passage.
7. Read common clinical data formats (CSV, Excel, SPSS, Stata, REDCap-style exports) and reshape data between wide and long layouts.

8. Use an AI coding assistant with structured prompts, and verify its code by checking denominators, ranges, duplicates and direction of comparisons.
9. Prepare tabular data using a dictionary, explicit cleaning rules, assertions, a cleaning log and a documented missing-data strategy.
10. Conduct and interpret univariate, bivariate and introductory multivariable analyses, including Welch tests, chi-square tests, correlation, linear regression and logistic regression.
11. Report effect estimates, 95% confidence intervals, p-values, assumptions, diagnostics and clinical relevance without confusing association with causation or odds with risk.
12. Generate reproducible tables, figures and reports, write a structured discussion linked to verified literature and frameworks, and disclose and verify AI assistance.

Course architecture and teaching approach

Each day produces inputs for the next. Day 1 defines the question, evidence plan and framework. Day 2 turns the required variables into an analysis-ready dataset. Day 3 estimates associations and evaluates uncertainty. Day 4 converts those outputs into a reproducible report and a calibrated discussion.

Day	Capability at exit	Main product carried forward
1. Research questions to evidence map	Explain what is being asked, which design and variables fit, and how sources will be verified	Question, hypotheses, framework map, search log, literature matrix and annotated bibliography entry
2. Data preparation and description	Read common formats, use an AI assistant safely, and defend every row and variable in the analysis set	Clean data, cleaning log, analysis-set flow, missing-data limitation, univariate pack
3. Estimation and regression	Choose an introductory method from outcome, exposure and design; interpret effects and uncertainty	Bivariate results, adjusted linear and logistic models, diagnostic notes
4. Reporting and reproducibility	Regenerate and communicate an auditable report without overstating the evidence	Mini report, results tables, figure, discussion, claim audit and handover package

Teaching methods

- Short demonstrations followed immediately by guided practice in pairs or teams of two to three.
- Predict, run, inspect: participants say what code should do before running it, and check the output afterwards.
- One visible research thread across slides, notebooks, worksheets and debriefs.
- Completion criteria for every exercise. Worked solutions are used for feedback, not copying.
- Bounded AI tasks with explicit input, permitted use, expected output and human checks. No AI-generated references or unverified numbers are accepted.
- Frequent interpretation practice in plain clinical language, including the difference between statistical precision and clinical importance.
- Python and R notebooks are line-for-line parallel, so the room follows one curriculum whichever language a participant uses.

Pre-course preparation

When	Activity
Two weeks before	Participants receive the setup guide and course folder; facilitators confirm institutional policy on Copilot and NotebookLM
One week before	60-minute online setup clinic: install VS Code, Python or R, Zotero and Quarto; sign in to GitHub Copilot; run Notebook 00
By Day 1	Notebook 00 prints *Setup check passed*. Anyone who

When	Activity
	cannot install software has a working Google Colab copy.

Detailed timetable

Breaks every day: 10:00 to 10:15, lunch 11:45 to 12:45, 14:15 to 14:30. Each block is 90 minutes.

Day 1: Research questions, design, literature and frameworks

Time	Focus	Learning and practice	Participant output
08:30 to 10:00	Questions and hypotheses	Clinical observation, PICO/PECO, H0/H1, FINER, estimand preview	Focused question and matched hypotheses
10:15 to 11:45	Design and frameworks	Inference limits, variable roles, conceptual and theoretical maps, COM-B	Variable and framework map
12:45 to 14:15	Search and Zotero	Concept blocks, search log, Zotero collections, metadata checks, PDF annotation	Traceable search and annotated records
14:30 to 16:00	NotebookLM synthesis	Zotero to NotebookLM workflow, bounded questions, five checks, matrix and bibliography	Verified three-source mini-review

Day 2: Data formats, AI-assisted coding, cleaning and description

Time	Focus	Learning and practice	Participant output
08:30 to 10:00	Notebooks, formats and AI coding	VS Code, kernels, tidy data, dictionaries; CSV, Excel, SPSS, Stata, REDCap; wide and long; GitHub Copilot and the five-part prompt	Working notebooks; audited AI code; first AI-use log entry
10:15 to 11:45	Data cleaning	Audit, standardise, validate, deduplicate, derive; assertions; cleaning log	Clean file and cleaning log
12:45 to 14:15	Missing data	MCAR, MAR, MNAR; clues in the data; complete-case analysis set and flow	Documented analysis-set decision and limitation
14:30 to 16:00	Univariate description	Summaries and graphs by variable type; denominators; descriptive language	Univariate evidence pack

Day 3: Estimation, bivariate analysis and regression

Time	Focus	Learning and practice	Participant output
08:30 to 10:00	Estimation and method choice	Estimand, effect scale, CI, p-value, method selection	Analysis plan with H0/H1 and effect measure
10:15 to 11:45	Bivariate analysis	Welch t-test, chi-square, correlation (demonstration), effect sizes	Mean difference and difference in proportions with 95% CIs

Time	Focus	Learning and practice	Participant output
12:45 to 14:15	Linear regression	Crude and adjusted models, reference levels, confounding, diagnostics	Adjusted mean difference and diagnostic note
14:30 to 16:00	Logistic regression	Probability, odds and OR; adjusted model; predicted probabilities	Adjusted OR and three-sentence interpretation

Day 4: Reproducible reporting, discussion and capstone

Time	Focus	Learning and practice	Participant output
08:30 to 10:00	Reproducible workflow	Restart and Run All, relative paths, assertions, session record, AI-use record	Checked end-to-end notebooks
10:15 to 11:45	Tables, figures and report	Table 1, focused Table 2, 300-dpi figure with text description, Quarto report	Results pack; rendered HTML and Word report
12:45 to 14:15	Structured discussion	D/L/F/A/R; limitations with direction; calibrated language; claim and evidence audit	Discussion matrix and audited discussion
14:30 to 16:00	Capstone and feedback	Report completion, peer review against the rubric, handover	Auditable mini research package

Session objectives and exercise sequence

Session	Exercise	Objectives	Completion evidence
Day 1, block 1	1A: Frame your clinical question	Convert an observation into PICO/PECO, state H0/H1 and revise once using FINER	One question with hypotheses that refer to the same population and contrast
Day 1, block 2	1B: Variable and framework map	Separate exposure, outcome, confounder, mediator and collider; map constructs to a framework	One-page map with measured and unmeasured constructs and one limitation
Day 1, block 3	Search and Zotero practicum	Build concept blocks, run and record a search, import and verify three records, annotate a PDF	Search log entry, three verified Zotero records, linked PDF notes
Day 1, block 4	1C: Literature mini-set	Move sources to NotebookLM, ask bounded questions, audit with the five checks	Three matrix rows, one annotated bibliography entry, verification status
Day 2, block 1	Notebooks 00 and 01; Exercises 0, 1A, 1B	Run a notebook; read five formats; prompt Copilot; audit AI-generated code	Both notebooks run; written judgement on the AI result; AI-use log entry
Day 2, block 2	2A: Audit and defend a cleaning rule	Standardise a category; implement and log rules; write a four-sentence rule defence	Clean file, cleaning log, passing assertions
Day 2, block 3	2B part 1: Who did we lose?	Tabulate missingness; reason about mechanisms; build the analysis set; compare included and	Flow table and a limitation linked to its consequence

Session	Exercise	Objectives	Completion evidence
		excluded	
Day 2, block 4	2B part 2: Univariate evidence pack	Choose summaries and graphs by type; check denominators; describe without causal language	One table, two figures, five-sentence interpretation
Day 3, block 1	3.0: Choose the method	Specify estimand and contrast before choosing a method	H0/H1, effect measure, method justification
Day 3, block 2	3A: Defend a bivariate analysis	Welch test for SBP by diabetes; chi-square for uncontrolled BP by diabetes	Mean difference and difference in proportions with 95% CI, p-values, one limitation each
Day 3, block 3	3B: Fit, check and interpret a linear model	Crude and adjusted models; per-10-point interpretation; residual diagnostics	Two-model comparison, adjusted estimate, diagnostic note
Day 3, block 4	3C: Defensible logistic interpretation	Adjusted OR; average predicted probabilities; odds versus risk	Three-sentence interpretation
Day 4, block 1	Reproducibility audit	Restart and Run All; resolve order and path problems; record environment	Clean runs and session information file
Day 4, block 2	4A and results-pack practicum	Rewrite weak prose; build Table 1, Table 2, figure; render the report	Results paragraph and rendered report files
Day 4, block 3	Discussion and claim audit	D/L/F/A/R matrix; limitations with direction; audit every claim	Discussion matrix and labelled discussion
Day 4, block 4	Capstone and peer review	Integrate the four days; review another team with the rubric	Final report, runnable notebook, evidence files, next-study plan

Assessment and capstone

Assessment is formative and product-based. Facilitators assess whether decisions are aligned, visible and verified, not whether participants obtain a preferred p-value. Participants may use solution notebooks after attempting an exercise but must explain their own choices in debriefs.

Assessment point	Evidence	Feedback focus
Day 1 exit	Question, hypotheses, framework map, verified literature mini-set	Alignment, inference boundary, traceability, measured and unmeasured constructs
Day 2 exit	Clean data, cleaning log, analysis-set flow, univariate pack, AI-use log	Reproducible rules, denominators, missing-data limitations, suitable summaries, AI verification
Day 3 exit	Bivariate and adjusted model outputs with interpretations	Contrast direction, units, reference categories, effect scale, uncertainty, diagnostics, non-causal wording
Day 4 capstone	Mini report and handover package	Rubric alignment, Run All success, discussion calibration, AI verification

Capstone brief. In teams of two or three, use the synthetic dataset or an instructor-approved equivalent to produce a two-page-equivalent mini research report. State the question and hypotheses; provide three verified literature-matrix entries and one annotated bibliography entry; include a framework map, cleaning decisions and analysis-set flow; provide Table 1, one main results table and one figure; finish with a D/L/F/A/R discussion and an AI-use and verification disclosure. Submit the notebook and all files needed to regenerate the reported outputs.

Criterion	Weight	Full-credit evidence
Question and framework	15	Focused question, matched hypotheses, coherent variable map, explicit inference boundary
Literature	15	Traceable search, verified sources, useful annotations, accurate source-to-claim links
Data and analysis	25	Documented cleaning, suitable models, diagnostics, missing-data decision and its limits
Results and discussion	25	Accurate estimates and uncertainty, clear table and figure, structured and proportionate interpretation
Reproducibility and AI use	20	Run All success, source files, output regeneration, disclosure and verification record

Suggested interpretation: 90 to 100 exemplary and auditable; 75 to 89 secure with minor revisions; 60 to 74 developing; below 60 incomplete alignment or verification. Teams receive feedback against evidence, not ranking.

Facilitation and learner support

- Before each demonstration, state the research decision and the expected output. Never present code as unexplained commands.
- Pair a more confident computer user with a less confident one, and rotate who types and who explains.
- Let participants adapt working code rather than type from memory. The workbook has a code-pattern reference for both languages.
- Run one curriculum. Python and R users sit together; the notebooks are parallel and produce identical numbers.
- Use large projection, downloadable files and scheduled breaks. Provide text descriptions for figures and do not rely on colour alone.
- Use the fallback pack if accounts, internet or software fail: Google Colab, printed source excerpts, screenshots, reference datasets and precomputed outputs preserve the learning objectives.

Setup and materials

Component	Requirement
Computer	Laptop with permission to install software; one per participant preferred, one per pair workable
Editor	Current VS Code with the Python, Jupyter and GitHub Copilot extensions (Quarto extension recommended)
Python route	Python 3.11 or newer; packages in requirements.txt (numpy, pandas, scipy, statsmodels, matplotlib, seaborn, openpyxl, pyreadstat, tabulate, ipykernel)

Component	Requirement
R route	R 4.3 or newer; run install_r_packages.R (tidyverse, broom, knitr, IRkernel)
Reporting	Quarto 1.4 or newer for the report and slides
AI accounts	GitHub account with Copilot (free plan is enough); NotebookLM only where institutional policy allows. Printed-source fallback fully supported.
Literature	Zotero desktop and browser connector
Fallback	Google Colab (browser) for participants who cannot install software

Course folder

Folder	Contents
course_guide/	This outline, facilitator guide, participant workbook, setup guide (Word)
slides/	Four Quarto slide decks (.qmd source and rendered .html)
notebooks/python/, notebooks/r/	Participant notebooks 00 to 04
notebooks/solutions/	Worked solutions in both languages, for facilitators
data/raw/	Synthetic raw CSV and data dictionary (never edited)
data/formats/	The same patients as Excel, SPSS, Stata, REDCap-style and long-format files
data/reference/	Reference clean and analysis datasets (fallback for later days)
literature/	Search log, literature matrix and facilitator source-list templates
templates/	Word templates: worksheet, analysis plan, cleaning log, AI-use log, annotated bibliography, discussion matrix, peer review, capstone report
reports/	Quarto mini-report template and styling
outputs/	Tables, figures and logs written by the notebooks

Quality, ethics and boundaries

- All course patient records and associations are simulated. They must not be cited as clinical evidence or used to change care.
- Do not upload identifiable patient data, confidential manuscripts, restricted full text or institutional secrets to public AI services. Follow local governance and copyright rules.
- AI may suggest search terms, explain or draft code, compare user-supplied passages and critique wording. Researchers remain responsible for sources, data decisions, statistical choices and claims.
- A citation supplied by AI is a verification task, not evidence. Verify the entity, passage, context, numbers and source record.
- AI-generated code is checked like any other code: run it, check denominators, ranges, duplicates and the direction of comparisons.
- Do not invent missing references, data, results or approvals. Mark unresolved items and get facilitator or methods review.
- Complete-case analysis is a teaching starting point, not a universal solution. Multiple imputation and other methods need further training.

References and further reading

- PICO/PECO, FINER and study design: Hulley SB, et al. *Designing Clinical Research*. Use the edition available through your institution.
- COM-B is introduced as an example theoretical lens for adherence, not as a tested mechanism in the synthetic data. Facilitators select and verify a primary COM-B source before delivery.
- Zotero documentation: zotero.org/support (adding items; PDF reader and note editor).
- NotebookLM: notebooklm.google. Product features and availability change; confirm before teaching.
- GitHub Copilot in VS Code: code.visualstudio.com/docs/copilot. Check current plan terms, including GitHub Education access.
- VS Code Jupyter notebooks: code.visualstudio.com/docs/datascience/jupyter-notebooks.
- Quarto: quarto.org.
- Python: pandas, SciPy, statsmodels, Matplotlib and seaborn documentation. R: tidyverse.org and the broom package documentation.
- Reporting guidelines: STROBE for observational studies, CONSORT for randomised trials, via equator-network.org.
- AI and research governance: current institutional policy, journal requirements, and privacy, data-protection and copyright law.

External product and methods resources should be reviewed by the facilitator before each delivery. The literature exercises intentionally contain no fabricated papers: facilitators choose real, relevant, open-access sources and verify each citation at the source.